RIGHT RESOLUTIONS BY FILTERED INJECTIVES: MORPHISM CASE

THE STACKS PROJECT AUTHORS

Editorial note. 'And so on' repeats the same displayed induction step mechanically (b55 precedent), transcribed as-is.

Editorial note. Source typo C[0] -> B[0] preserved verbatim.

Allowed standard prerequisites (editorial).

- Filtered injective objects and filtered acyclicity. lemma-injective-property-filtered-injective (same chapter) and homology-lemma-filtered-acyclic (cross-chapter), cited in-span; not carded. Supply the filtered-injectivity engine used at the induction step and the acyclicity link for the resolution.

Source reference. Original reference 05QH: focused-seeds/geometry/S9-der-filtered-injective-right-resolution-map/ref-tags.txt, source lines 5–5. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original lemma 05TR: sources/stacks-derived/pinned-original-source.tex, source lines 8113–8113. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Lemma 0.1. *Let $\mathcal{A}$ be an abelian category with enough injectives. Let $f : A \to B$ be a morphism of $Fil^f(\mathcal{A})$. Given filtered quasi-isomorphisms $A[0] \to I^\bullet$ and $B[0] \to J^\bullet$ where $I^\bullet, J^\bullet$ are complexes of filtered injective objects with $I^n = J^n = 0$ for $n < 0$, then there exists a commutative diagram*

$$\begin{array}{ccc} A[0] & \longrightarrow & B[0] \\ \downarrow & & \downarrow \\ I^\bullet & \longrightarrow & J^\bullet \end{array}$$

Proof. As $A[0] \to I^\bullet$ and $C[0] \to J^\bullet$ are filtered quasi-isomorphisms we conclude that $a : A \to I^0$, $b : B \to J^0$ and all the morphisms d_I^n, d_J^n are strict, see Homology, Lemma 05QH. We will inductively construct the maps f^n in the following commutative diagram

$$\begin{array}{ccccccccc} A & \xrightarrow[a]{} & I^0 & \longrightarrow & I^1 & \longrightarrow & I^2 & \longrightarrow & \ldots \\ {\scriptstyle f}\downarrow & & \downarrow{\scriptstyle f^0} & & \downarrow{\scriptstyle f^1} & & \downarrow{\scriptstyle f^2} & & \\ B & \xrightarrow{b} & J^0 & \longrightarrow & J^1 & \longrightarrow & J^2 & \longrightarrow & \ldots \end{array}$$

Date: stacks-project:derived.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Because $A \to I^0$ is a strict monomorphism and because J^0 is filtered injective, we can find a morphism $f^0 : I^0 \to J^0$ such that $f^0 \circ a = b \circ f$, see Lemma 05TR. The composition $d^0_J \circ b \circ f$ is zero, hence $d^0_J \circ f^0 \circ a = 0$, hence $d^0_J \circ f^0$ factors through a unique morphism

$$\mathrm{Coker}(a) = \mathrm{Coim}(d^0_I) = \mathrm{Im}(d^0_I) \longrightarrow J^1.$$

As $\mathrm{Im}(d^0_I) \to I^1$ is a strict monomorphism we can extend the displayed arrow to a morphism $f^1 : I^1 \to J^1$ by Lemma 05TR again. And so on. $\square$